Calendar — Fundamentals

Odd Days, Leap Years & the Day-of-the-Week Formula

Aptitude Grind

Why Calendar Problems Matter

Calendar questions test whether you can find the **day of the week** for a given date, or work backward/forward from a known day. They show up constantly in placement tests (TCS NQT, Infosys, Wipro, Capgemini) and in CAT/other reasoning sections, usually as 1–2 quick-scoring questions **if** you know the odd-days method cold. The entire topic rests on one idea: *the calendar repeats every 7 days, so only the remainder after dividing by 7 matters.*

1 Odd Days

Definition

In a given period, the number of days **more than complete weeks** is called the number of **odd days**.

For example, 17 days = 2 complete weeks (14 days) + 3 extra days $\Rightarrow$ 3 odd days. Only the odd-day count matters for finding the day of the week — full weeks never shift the day.

1.1 Odd Days in a Year

- 1 ordinary year = 365 days = 52 weeks +1 day $\Rightarrow$ **1 odd day**
- 1 leap year = 366 days = 52 weeks +2 days $\Rightarrow$ **2 odd days**

1.2 Odd Days in Centuries

Century Odd-Day Table — memorize this

Period	Odd Days
100 years	5
200 years	3
300 years	1
400 years	0

Why 400 years gives 0: 100 years = 76 ordinary +24 leap years = $(76 \times 1 + 24 \times 2) = 124$ odd days = 17 weeks +5 days $\Rightarrow$ 5 odd days. So 400 years = $(5\times4)+1$ extra leap day (400 itself is a leap century) = $21 \equiv 0 \pmod 7$.

★ Pattern

Every 4th century (400, 800, 1200, 1600, 2000, 2400, ...) has **0 odd days** — meaning the calendar of year Y is identical to the calendar of year $Y + 400$. This is why 2000 and 2400 have the exact same calendar.

2 Leap Years

Leap Year Rule

1. Every year divisible by 4 is a leap year — **unless** it is a century year (ends in 00).
2. A century year is a leap year only if it is divisible by **400**.

Quick Classification

- **Leap:** 1948, 2004, 1676, 2000, 1600, 2400 (divisible by 4; century years also divisible by 400)
- **Not leap:** 2001, 2002, 2003, 2005 (not divisible by 4); 1800, 1900, 2100, 2300 (century years *not* divisible by 400)

Common Mistake

Students often forget the century exception and assume any year divisible by 4 is automatically a leap year. **1900 is divisible by 4 but is NOT a leap year** (not divisible by 400). Always check century years separately.

3 Mapping Odd Days to a Day of the Week

Once you compute the total number of odd days for a period (taking the result mod 7), map it using:

Odd Days → Day Table

Odd Days	0	1	2	3	4	5	6
Day	Sun	Mon	Tue	Wed	Thu	Fri	Sat

This table is anchored on the fact that **1 January, 1 AD was a Monday** in the proleptic Gregorian calendar used for these problems, giving Sunday = 0 odd days as the baseline.

4 General Method: Finding the Day for Any Date

To find the day of the week on date D of month M, year Y:

Step-by-Step Method

1. Split $Y - 1$ (the reference is "up to the end of the previous year") into:
   - Number of 400s $\to$ always 0 odd days
   - Remaining 100s (0 to 3) $\to$ 5, 3, or 1 odd days respectively
   - Remaining years $\to$ (ordinary years $\times 1$) + (leap years $\times 2$)
2. Add the odd days contributed by the **months elapsed so far in year** Y (Jan 1 to the last day of the previous month), remembering to use 29 for February in a leap year.
3. Add the day number D of the current month.
4. Sum everything, take mod 7, and read off the day from the Odd Days $\to$ Day table.

Worked Example — 15 August 2010

Step 1: Years up to 2009.

- 1600 years $\to$ 0 odd days; 400 years (1600–1999... use nearest 400 block) $\to$ 0 odd days
- Remaining 9 years (2001–2009): 2004 and 2008 are leap $\Rightarrow$ 2 leap + 7 ordinary $= (2 \times 2) + (7 \times 1) = 11 \equiv 4$ odd days

Step 2: Months elapsed (Jan–Jul 2010) + 15 days of Aug.

$$31 + 28 + 31 + 30 + 31 + 30 + 31 + 15 = 227 \text{ days} = 32 \text{ weeks} + 3 \text{ days} \Rightarrow 3 \text{ odd days}$$

Step 3: Total.

$$0 + 0 + 4 + 3 = 7 \equiv 0 \pmod 7 \Rightarrow \textbf{Sunday}$$

15 August 2010 was indeed a **Sunday**.

★ Faster Alternative

For placement-test speed, most people skip the "years since year 1" method entirely and instead work **relative to a known anchor date** (e.g., "1 Jan 2000 was a Saturday" or "1 Jan 1 was Monday") using only the years *between* the anchor and the target date. This is faster and less error-prone — see `02-tricks-and-shortcuts.tex` for the anchor-date and month-code methods.

5 Key Takeaways

- Everything reduces to counting **odd days** and taking the result mod 7.
- Memorize: 100y $\to$ 5, 200y $\to$ 3, 300y $\to$ 1, 400y $\to$ 0 odd days.
- Leap year = divisible by 4, except century years, which need divisibility by 400.
- Always double check whether February in the target year has 28 or 29 days.